

EE100

Basic Electrical Engineering

Module – 04: Electrical Machines

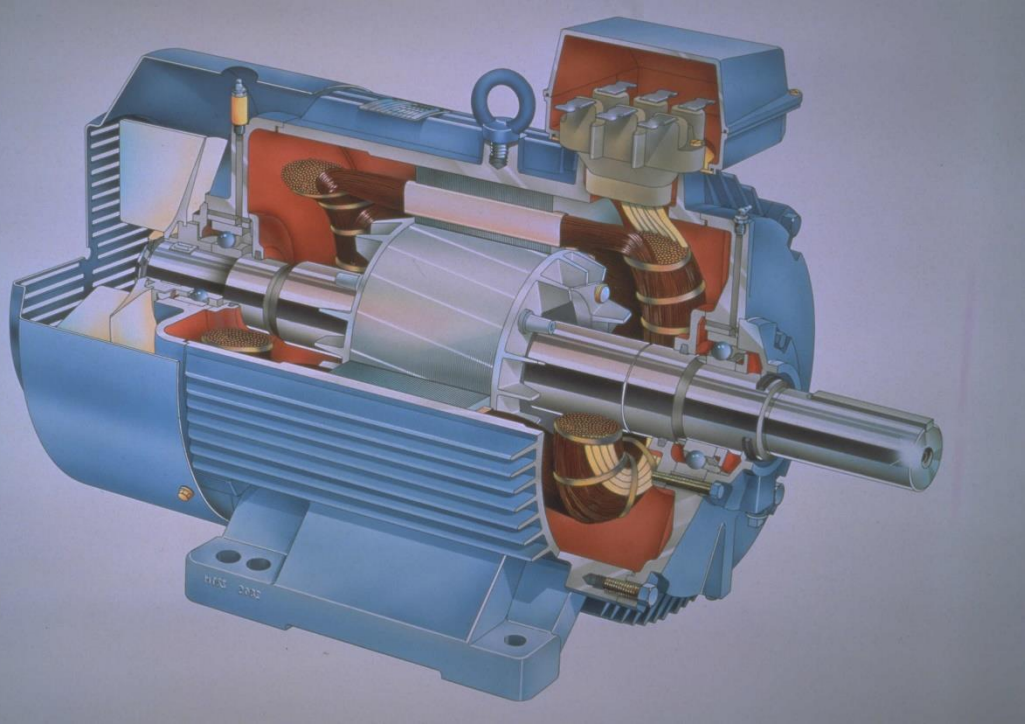
Three phase Induction Motor



**Dept. of Electrical Engineering,
C V RAMAN GLOBAL UNIVERSITY
BHUBANESWAR, ODISHA**



Three Phase Induction Motor



- ❖ Three-phase induction motors are the most common and frequently encountered machines in industry
 - Simple design, rugged, low-price, easy maintenance
 - Wide range of power ratings: fractional horsepower to 10 MW
 - Run essentially as constant speed from no-load to full load
 - Its speed depends on the frequency of the power source
 - ❑ not easy to have variable speed control
 - ❑ requires a variable-frequency power-electronic drive for optimal speed control

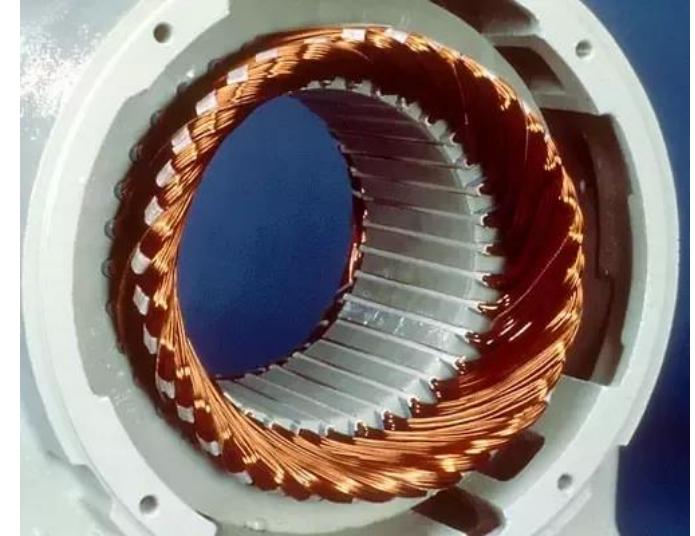


Three Phase Induction Motor

❖ An induction motor has two main parts

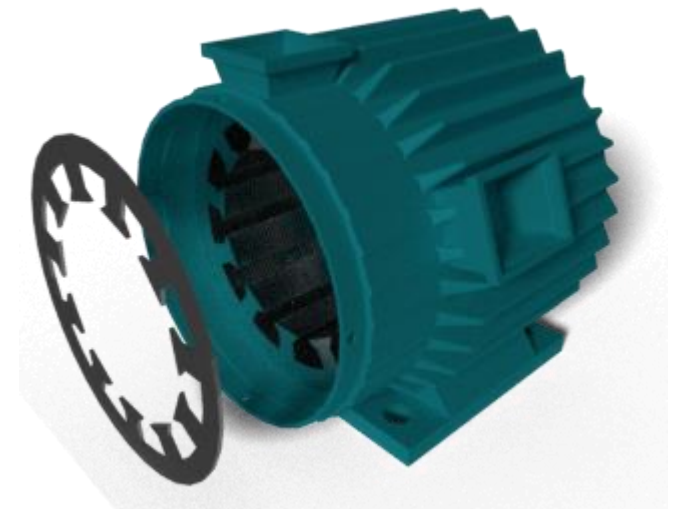
➤ a stationary stator

- ❑ consisting of a steel frame that supports a hollow, cylindrical core
- ❑ core, constructed from stacked laminations, having a number of evenly spaced slots, providing the space for the stator winding



➤ a revolving rotor

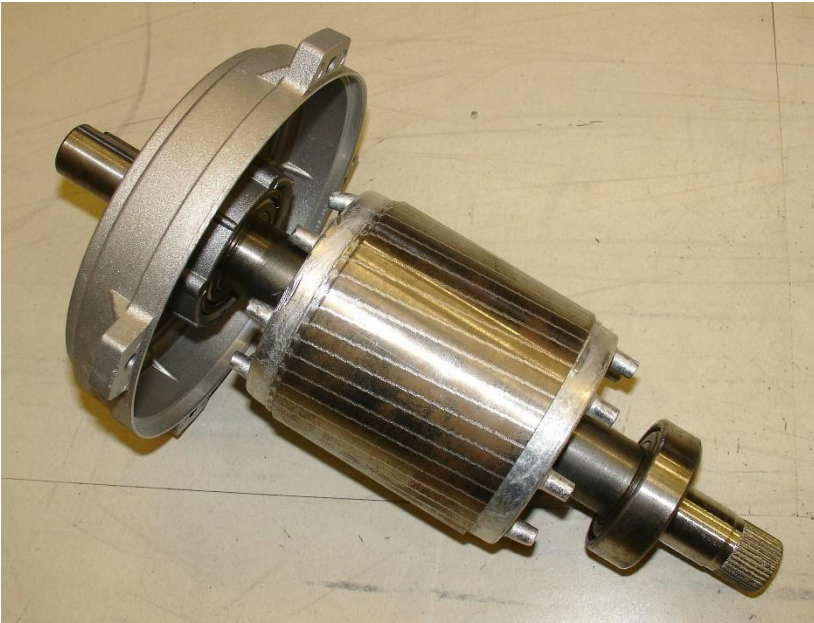
- ❑ composed of punched laminations, stacked to create a series of rotor slots, providing space for the rotor winding
- ❑ one of two types of rotor windings
 - ❑ conventional 3-phase windings made of insulated wire (**wound-rotor**) » similar to the winding on the stator
 - ❑ aluminum bus bars shorted together at the ends by two aluminum rings, forming a squirrel-cage shaped circuit (**squirrel-cage**)





Three Phase Induction Motor

Two basic design types depending on the rotor design



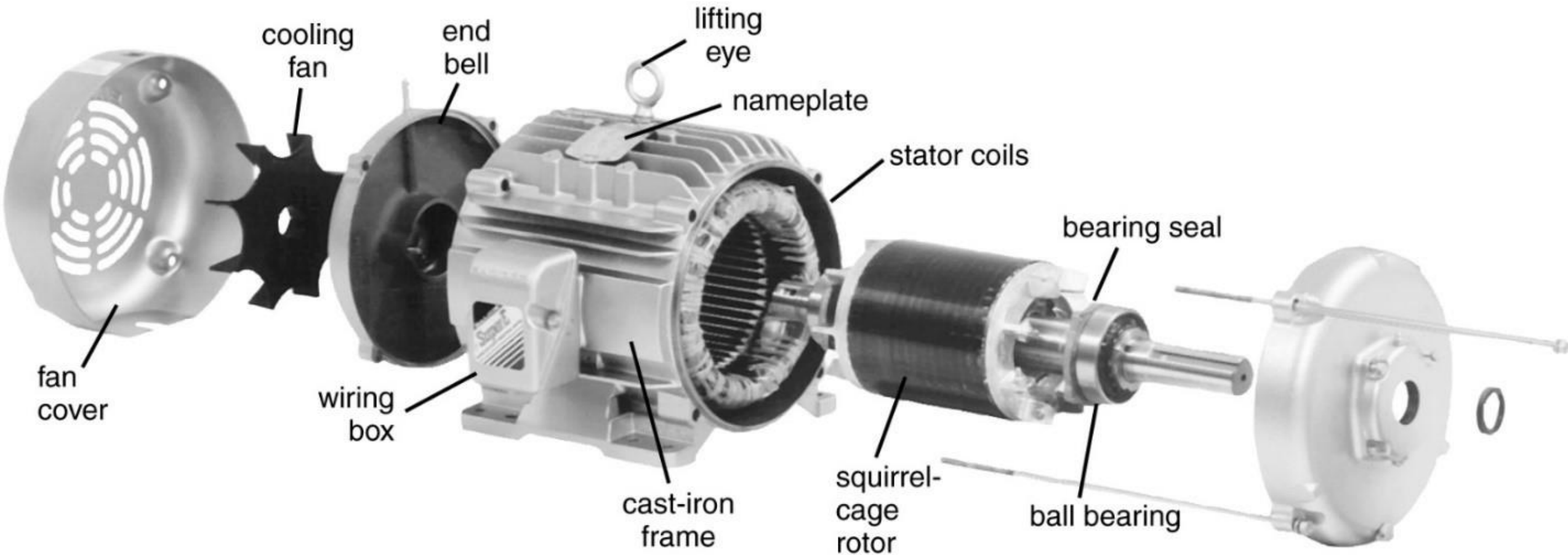
Squirrel-cage: conducting bars laid into slots and shorted at both ends by shorting rings.



Wound-rotor: complete set of three-phase windings exactly as the stator. Usually Y-connected, the ends of the three rotor wires are connected to 3 slip rings on the rotor shaft. In this way, the rotor circuit is accessible.



Three Phase Induction Motor



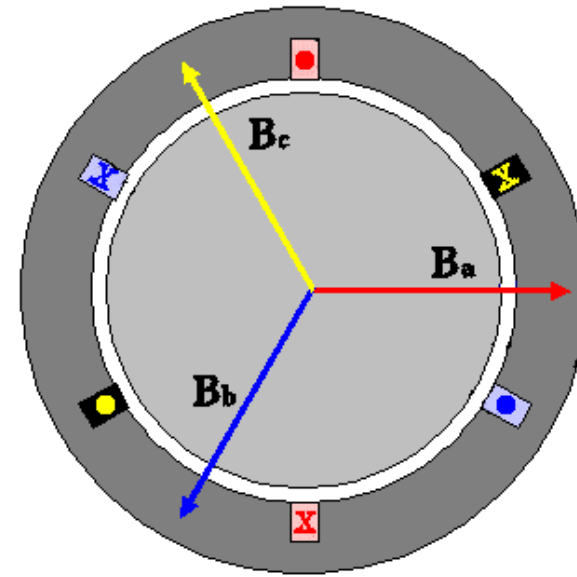
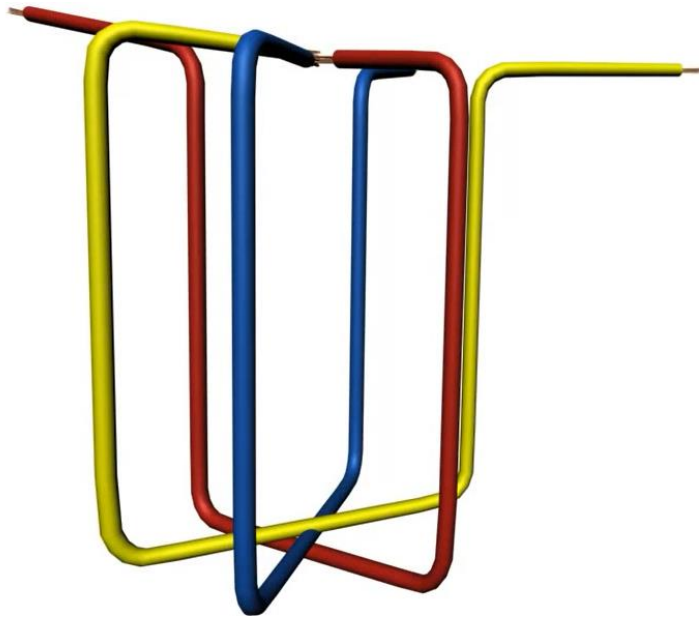


Rotating Magnetic Field

- ❖ Balanced three phase windings, i.e. mechanically displaced 120 degrees from each other, fed by balanced three phase source
- ❖ A rotating magnetic field with constant magnitude is produced, rotating with a speed

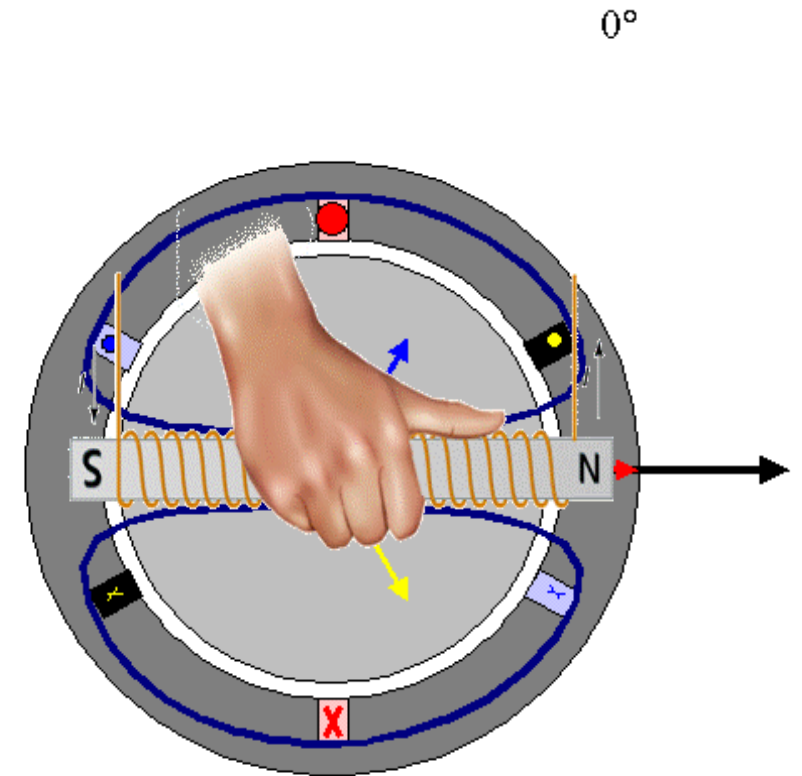
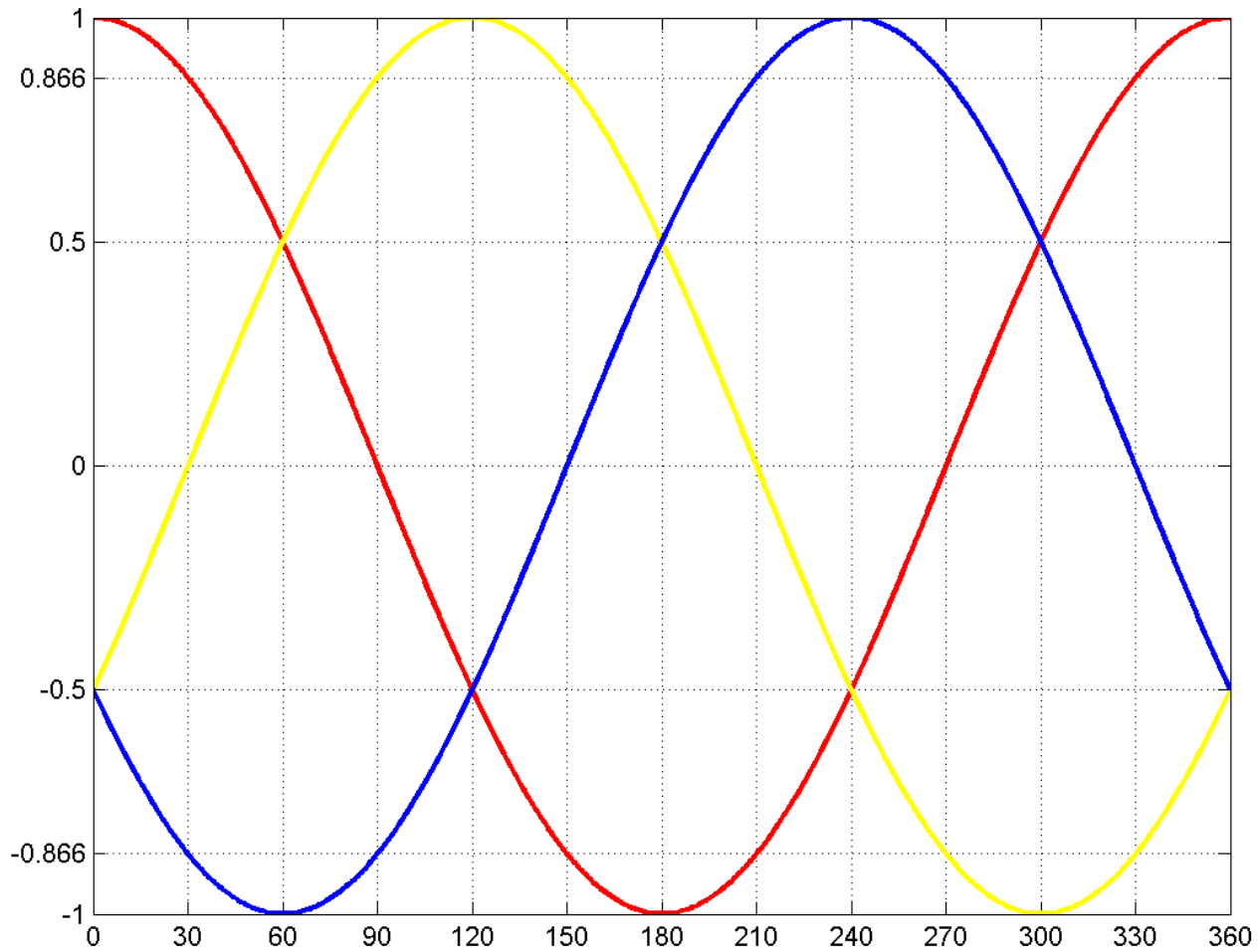
$$n_{sync} = \frac{120 f_e}{P} \text{ rpm}$$

➤ Where, f_e is the supply frequency, P is the no. of poles and n_{sync} is called the synchronous speed in rpm (revolutions per minute)



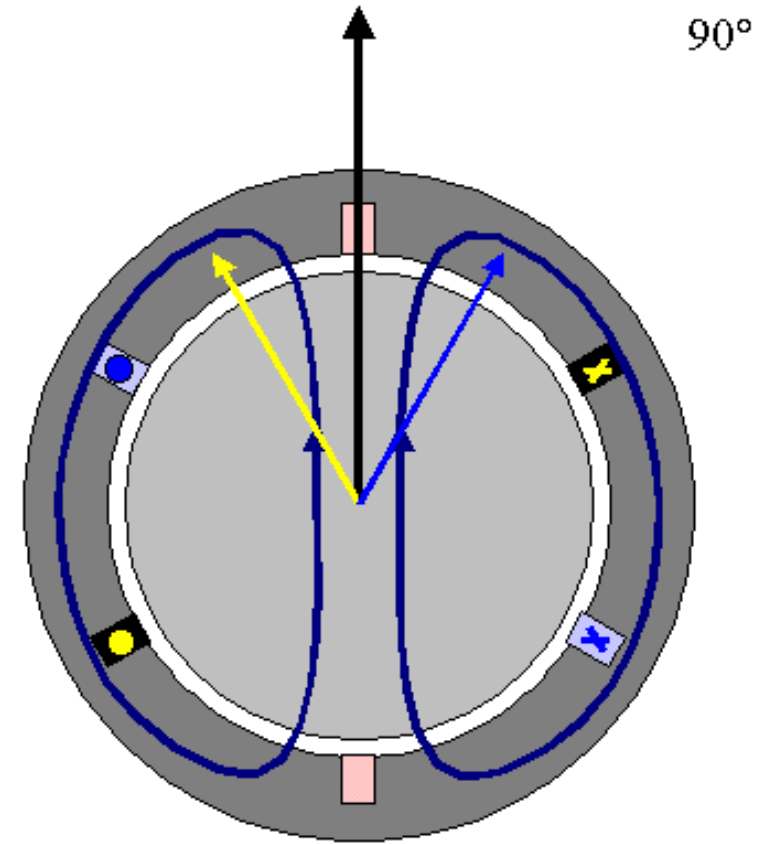
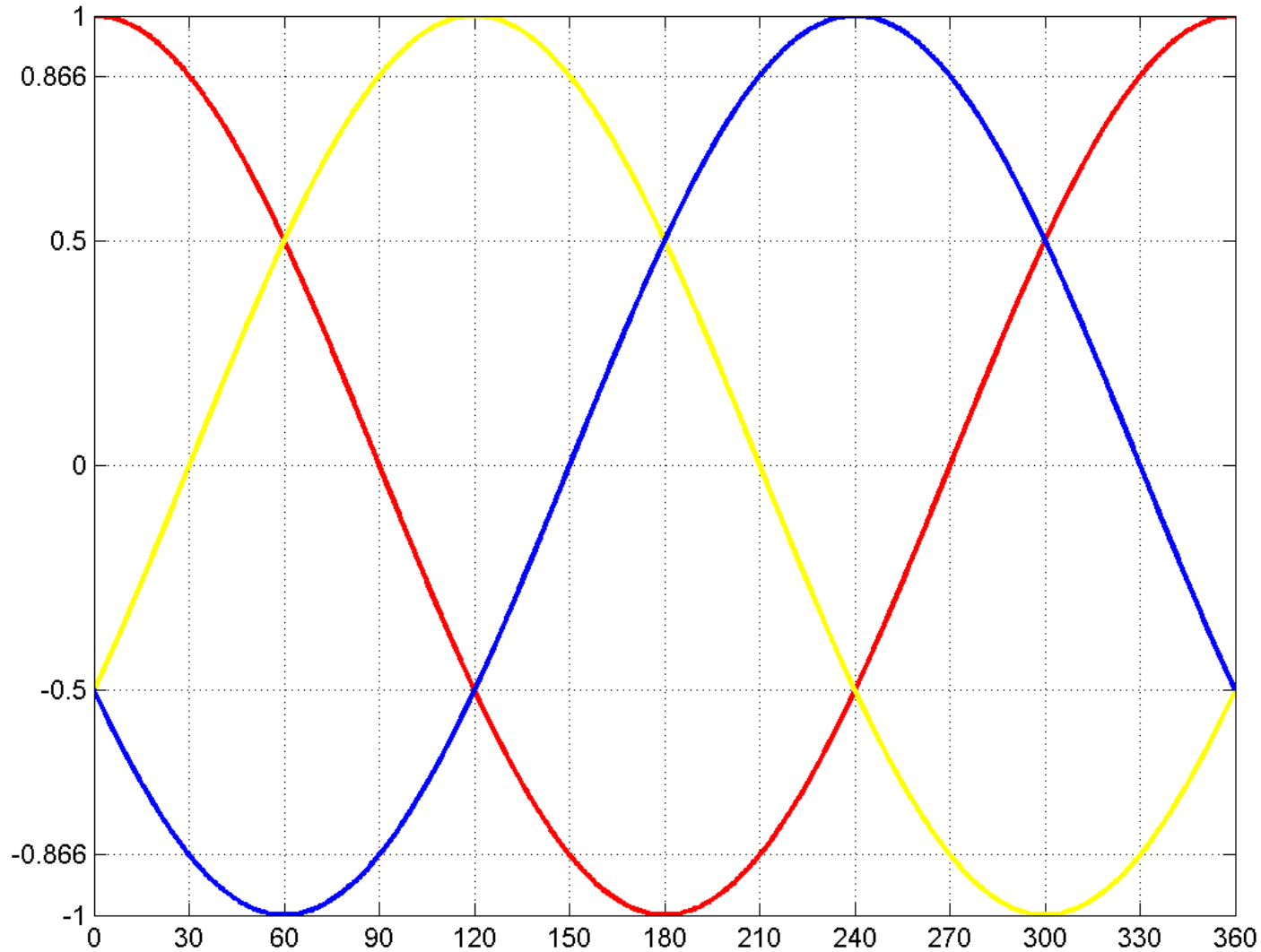


Rotating Magnetic Field



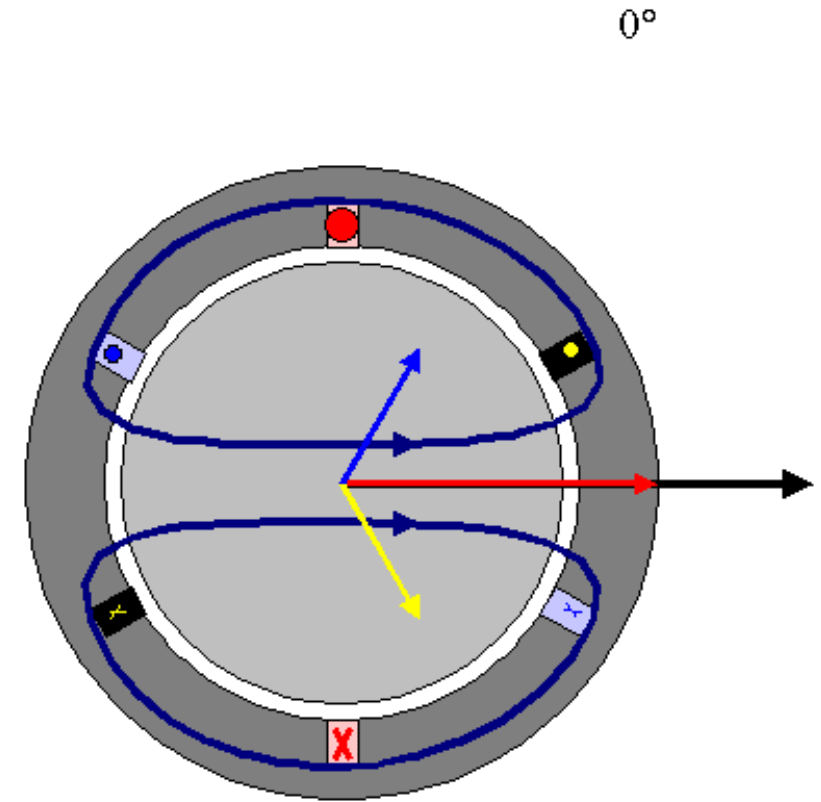
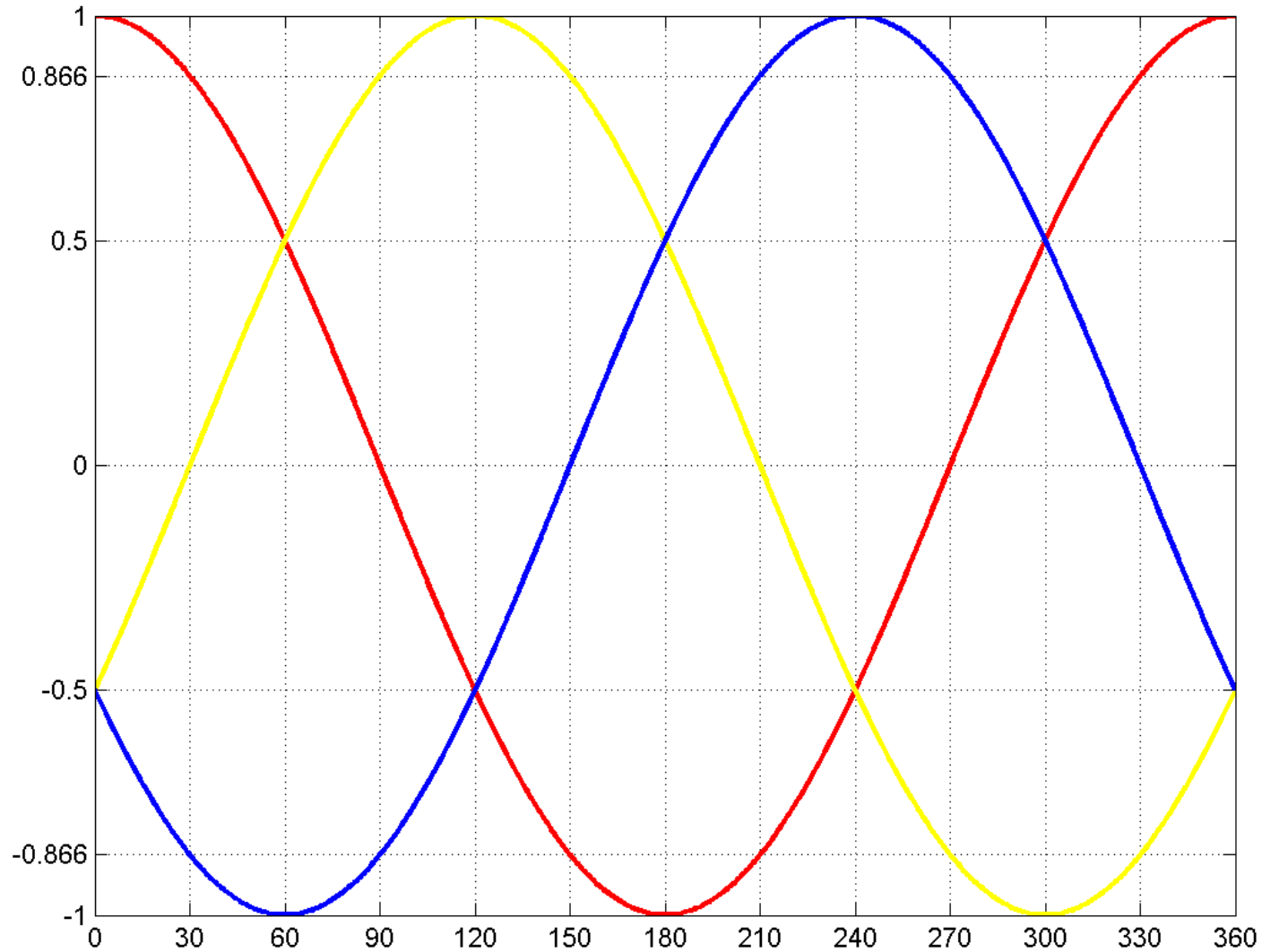


Rotating Magnetic Field



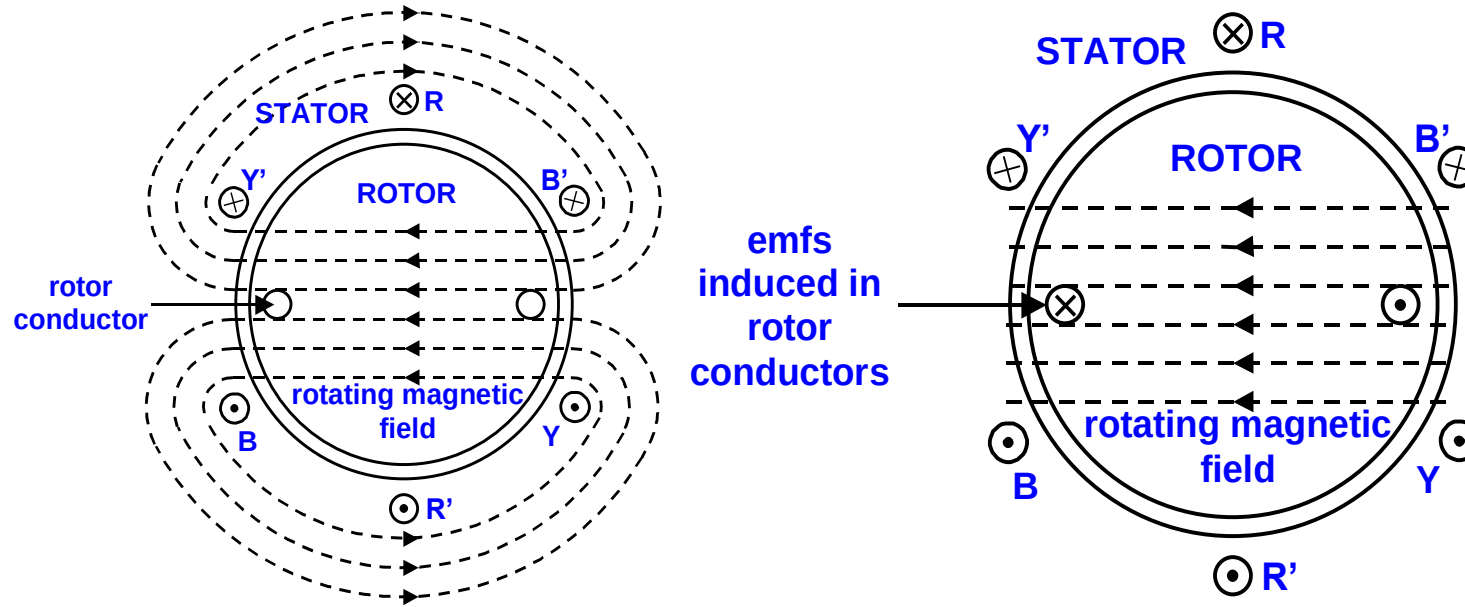


Rotating Magnetic Field





Rotating Magnetic Field



Consider a simple rotor, with one short circuited coil, inserted within the stator:

- ❖ Initially, the rotor is stationary.
- ❖ The moment the stator supply is switched on currents start to flow and the rotating magnetic field is established.
- ❖ The relative motion between the moving field and the stationary rotor conductors induces emf in the stationary rotor conductors (in accordance with Faraday's Law)



Three Phase Induction Motor

- ❖ Current start flowing in the conductors as they are short circuited by the end rings.
- ❖ These currents create their own magnetic fields, which interact with the rotating stator field to produce forces on the individual conductors and a net rotor torque
- ❖ The rotor starts to accelerate lowering the relative speed between the rotating field and rotor conductors.
 - This reduces the induced emfs, conductor currents and subsidiary magnetic fields;
 - thus decreasing the forces on the conductors and electrical torque on the rotor.
- ❖ The rotor continues to accelerate until the electrical torque exactly equals the mechanical load torque on the shaft.
 - At this point the rotor is running at a speed slightly slower than the rotating field.
 - This small difference in speed is needed.
 - In order to create an electrical torque there must be some distortion of the net field, which will only happen when currents flow in the rotor conductors.
 - These currents depend on emfs being induced in the conductors, which in turn depend on there being a difference between the speed at which the conductors rotate and that of the rotating magnetic field.



Three Phase Induction Motor

- ❖ This difference in speed is expressed as a ratio known as the (per unit) slip.
- ❖ Remembering that the rotational speed of magnetic field is known formally as the synchronous speed, the slip is defined as

$$\text{Slip} = \frac{(\text{synchronous speed, } n_s) - (\text{actual rotor speed, } n)}{(\text{synchronous speed, } n_s)}$$

- ❖ For most machines the value of the slip varies between around 0.01 on no-load, (when the only torque required is to overcome friction at the bearings) and 0.10 at full load.

Applications of Three Phase Induction Motor

- | | |
|--|---|
| ☐ Lifts | ☐ Driving lathe machines |
| ☐ Cranes | ☐ Crushers |
| ☐ Hoists | ☐ Oil extracting mills |
| ☐ Large capacity exhaust fans | ☐ Textile and etc. |



Thank You !